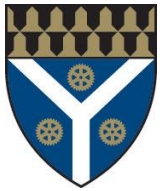




Probabilistic Forecasting of S&P 500 Stocks Using Conditioned Normalizing Flows

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INTRODUCTION

Forecasting the behavior of financial markets remains a long-standing problem in statistics, machine learning, and economics.

Traditional and industry standard forecasting methods often rely on assumptions of linearity, gaussianity of returns distribution, and homoscedasticity which frequently don't hold true in dynamic, public markets. Deep learning attempts have improved forecasts, but still don't learn the full return distribution from the data, again, relying on gaussianity.

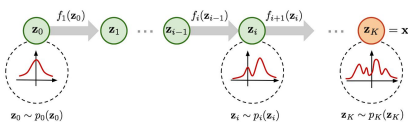
To solve this problem the use of CNFs is proposed to try to learn a target distribution X exactly, without any assumptions about the distribution's shape.

Key Contributions

- > Development a CNF model with an LSTM encoder for time-series forecasting of S&P 500 stocks
- > Implemented ARIMA and standard Gaussian LSTM forecast models to serve as performance baselines
 - > Evaluates the model against baselines using probabilistic metrics including Continuous Ranked Probability Score (CRPS) and visual forecast plots.
- > Constructed a daily-resolution S&P 500 dataset with aligned macroeconomic indicators and Fourier features, processed using a reproducible pipeline

BACKGROUND

Normalizing Flows (NFs) are generative models which learn a series of invertible transformations from a simple latent distribution to a target.



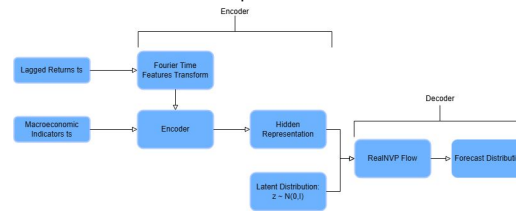
Conditioned Normalizing Flows (CNFs) extend the behavior of NFs to be conditioned on exogenous variables. If we let $f = f_1 \circ f_2 \circ \dots \circ f_K$ then the CNF conditions lets us condition f on context variables: $Z_i = f(X_i | C_i)$.

METHODOLOGY

The training dataset of 600,000 points, spanning the period from January 2019 to January 2024, was created by pulling S&P 500 equity prices, VIX levels, and the 13-week treasury bill rate as context variables. Equity prices, were standardized them to log returns in preprocessing.

The architecture consists of two main components:

1. A Real NVP (Non Volume Preserving) flow decoder that learns an invertible transformation to a target distribution
2. A two layer LSTM as a recurrent encoder that passes some hidden representation into the decoder.



The model was trained using maximum likelihood estimation with the objective function:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log p_Z(f_\theta(X_i | C_i)) + \log \left| \det \frac{\partial f_\theta}{\partial X_i} \right|.$$

This objective minimizes the negative loss likelihood of the transformed samples, $f_\theta(X_i | C_i)$, under a standard Gaussian.

RESULTS

To test the model CRPS scores, coverage intervals between the CNF and baselines, and forecast plots were made for visual explainability. The CNF model outperforms the Gaussian LSTM and ARIMA baselines in CRPS metric, indicating slightly better calibration of its forecast distributions. The CNF model's empirical coverage for its 80% predictive exhibits larger undercoverage than the overcoverage shown in the baselines. This means the CNF misses the groundtruth more often and is possibly overconfident in its forecasts, but the predictions are more precise in where the model thinks outcomes will land.

Model	Mean CRPS	Standard Deviation
CNF (ours)	0.0138	0.0112
Gaussian LSTM	0.0153	0.0172
ARIMA	0.0165	0.0194

Figure: Mean CRPS and standard deviation scores across 5 randomized seeds and 100 windows

Model	Nominal Coverage	Empirical Coverage
CNF (ours)	80%	71.59%
Gaussian LSTM	80%	86.88%
ARIMA	80%	87.44%

Figure: Empirical 80% interval coverage across 5 randomized seeds and 100 windows

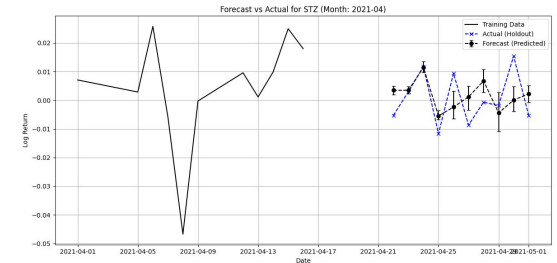


Figure: Sample forecast plot visualization configured from CNF output

DISCUSSION

The model's CRPS scores combined with the empirical coverage compared to industry baselines suggest sharp forecast accuracy, but also overconfident uncertainty estimates. This may be due to overfitting to the macroeconomic context features. Unlike the baselines, which because of their gaussianity assumptions can be resilient to sudden changes in context variable shifts, the CNF will always try to adapt its distribution output. While this is an advantage, it comes with the risk of overreacting to noise, as may be the case here.

Some future work may include:

1. Incorporating cross-series attention mechanisms to learn market dependencies
2. Train the model using a flow matching algorithm. Flow matching is relatively simulation free and could be more efficient and scalable
 - a. Guided flow matching may further improve accuracy by using a reward mechanism from RL